Types KD, HKD and KDC Electronic Circuit Breakers with Interchangeable Trip Units

Order as individual components: Breaker Frame, Trip Unit, Rating Plug, Terminals.

Table 12-175. Types KD, HKD and KDC Electronic Circuit Breakers with Interchangeable Trip Units ①

Max. Cont. Ampere Rating at 40°C ②	Circuit Breaker Frame Only			Digitrip RMS 310 Trip Unit Only ③				Digitrip 310 Rating Plug Only			Standard Terminals Only See Page 12-107 for Optional Terminals
	Standard Interrupting Capacity 600 Vac Rated 35 kAIC at 480 Vac	High Interrupting Capacity 600 Vac Rated 65 kAIC at 480 Vac	Ultra High Interrupting Capacity Current Limiting 600 Vac Rated 100 kAIC at 480 Vac	Standard	Options			Ampere Rating	Fixed Rating Plug	Adjustable Rating Plug Ampere Rating	
				Adjustable Short Time Pickup with I^2t Short Delay Ramp	Independently Adjustable Short Time Pickup and Delay	Adjustable Short Time Pickup with I^2t Short Delay and Ground Fault Protection	Independently Adjustable Short Time Pickup and Delay and Ground Fault Protection				
	Catalog Number								Catalog Number		
3-Pole ④											
125	**KD3400F**	**HKD3400F**	**KDC3400F**	**KES3125LS**	**KES3125LSI**	**KES3125LSG**	**KES3125LSIG**	70 90 100 110 125	**1KES070T** **1KES090T** **1KES100T** **1KES110T** **1KES125T**	70/90/100/125 **A1KES125T1**	**TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤
250	**KD3400F**	**HKD3400F**	**KDC3400F**	**KES3250LS**	**KES3250LSI**	**KES3250LSG**	**KES3250LSIG**	70 100 125 150 160 175 200 225 250	**2KES070T** **2KES100T** **2KES125T** **2KES150T** **2KES160T** **2KES175T** **2KES200T** **2KES225T** **2KES250T**	125/150/200/250 **A2KES250T1**	 **TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA350K** ⑤
400	**KD3400F**	**HKD3400F**	**KDC3400F**	**KES3400LS**	**KES3400LSI**	**KES3400LSG**	**KES3400LSIG**	200 225 250 300 350 400	**4KES200T** **4KES225T** **4KES250T** **4KES300T** **4KES350T** **4KES400T**	200/250/300/400 **A4KES400T1** 200/240/260/280 **A4KES200T5** 320/340/360/380 **A4KES300T5**	**TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA350K** ⑤ **3TA400K** ⑥
4-Pole ⑦⑧											
125	**KD4400F**	**HKD4400F**	**KDC4400F**	**KES4125LS**	**KES4125LSI**	—	—	70 90 100 110 125	**1KES070T** **1KES090T** **1KES100T** **1KES110T** **1KES125T**	70/90/100/125 **A1KES125T1**	**TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤
250	**KD4400F**	**HKD4400F**	**KDC4400F**	**KES4250LS**	**KES4250LSI**	—	—	70 100 125 150 160 175 200 225 250	**2KES070T** **2KES100T** **2KES125T** **2KES150T** **2KES160T** **2KES175T** **2KES200T** **2KES225T** **2KES250T**	125/150/200/250 **A2KES250T1**	**TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA300K** ⑤ **TA350K** ⑤
400	**KD4400F**	**HKD4400F**	**KDC4400F**	**KES4400LS**	**KES4400LSI**	—	—	200 225 250 300 350 400	**4KES200T** **4KES225T** **4KES250T** **4KES300T** **4KES350T** **4KES400T**	200/250/300/400 **A4KES400T1** 200/240/260/280 **A4KES200T5** 320/340/360/380 **A4KES300T5**	**TA300K** ⑤ **TA300K** ⑤ **TA350K** ⑤ **TA350K** ⑤ **TA350K** ⑤ **4TA400K** ⑥

① See **Table 12-176** on **Page 12-99** for prices.
② Ampere rating is established by rating plug.
③ For ac use only.
④ 3-pole KES Trip Units are for use in 3-pole frames only.
⑤ Individually packed.
⑥ 3TA400K and 4TA400K terminal kits contain one terminal for each pole and one terminal cover.
⑦ Trip unit includes protected neutral pole. Use corresponding 3-pole trip unit if protected neutral pole is not required.
⑧ Fully rated neutral pole protection is standard. For 50% rated protection on neutral pole, add Suffix **E** to 4-pole trip unit Catalog Number.

Product Selection

Table 12-176. Types KD, HKD and KDC Electronic Circuit Breakers with Interchangeable Trip Unit Prices

Catalog Number	Price U.S. $	Catalog Number	Price U.S. $	Catalog Number	Price U.S. $	Catalog Number	Price U.S. $
A1KES125T1 A2KES250T1 A4KES400T1 HKD3400F HKD4400F		KES3250LSI KES3250LSIG KES3400LS KES3400LSG KES3400LSI		1KES090T 1KES100T 1KES110T 1KES125T 2KES070T		4KES225T 4KES250T 4KES300T 4KES350T 4KES400T	
KDC3400F KDC4400F KD3400F KD4400F		KES3400LSIG KES4125LS KES4125LSI KES4250LS		2KES100T 2KES125T 2KES150T 2KES160T		4TA400K — — —	
KES3125LS KES3125LSG KES3125LSI KES3125LSIG KES3250LS KES3250LSG		KES4250LSI KES4400LS KES4400LSI TA300K TA350K 1KES070T		2KES175T 2KES200T 2KES225T 2KES250T 3TA400K 4KES200T		— — — — — —	

Table 12-177. Types KDB and HKDB with Non-Interchangeable Trip Unit Suitable for Reverse Feed

Maximum Continuous Ampere Rating at 40°C	Number of Poles	Circuit Breaker Frame Including Digitrip RMS 310 Electronic Tip Unit Less Terminals and Rating Plug				Digitrip RMS 310 Rating Plug (Order as Separate Items)	
		L – Adjustable Long Delay Pickup (By Adjustable Rating Plug) S – Adjustable Short Delay Pickup with Fixed Short Delay Time (I^2t Response) or Adjustable Short Time (Flat Response) I – Adjustable Instantaneous Pickup by Setting Short Time Delay Time to Instantaneous G – Adjustable Ground Fault Pickup with Adjustable Ground Fault Delay (Flat Response)				Fixed	Adjustable
		LS	LSI	LSG	LSIG		
		Catalog Number				Catalog Number	
		Short Time Range Short Time Delay Ground Fault Pickup Ground Fault Delay					
		2 – 8 x I_n — — —	2 – 8 x I_n 0 – 300 ms — —	2 – 8 x I_n — Varies by Frame 0 – 500 ms	2 – 8 x I_n 0 – 300 ms Varies by Frame 0 – 500 ms		
Type KDB with Digitrip 310 Non-Interchangeable Trip Unit Suitable for Reverse Feed							
125	3	KDB3125FT33W	KDB3125FT32W	KDB3125FT35W	KDB3125FT36W	1KES070T 1KES090T 1KES100T 1KES125T	70/90/100/125 A1KES125T1
250	3	KDB3250FT33W	KDB3250FT32W	KDB3250FT35W	KDB3250FT36W	2KES070T 2KES090T 2KES100T 2KES125T 2KES150T 2KES160T 2KES175T 2KES200T 2KES225T 2KES250T	125/150/200/250 A2KES250T1
400	3	KDB3400FT33W	KDB3400FT32W	KDB3400FT35W	KDB3400FT36W	4KES200T 4KES225T	200/250/300/400 A4KES400T1
						4KES250T 4KES300T 4KES350T 4KES400T	200/240/260/280 A4KES200T5
							320/340/360/380 A4KES300T5

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Table 12-177. Types KDB and HKDB with Non-Interchangeable Trip Unit Suitable for Reverse Feed (Continued)

Maximum Continuous Ampere Rating at 40°C	Number of Poles	Circuit Breaker Frame Including Digitrip RMS 310 Electronic Tip Unit Less Terminals and Rating Plug				Digitrip RMS 310 Rating plug (Order as Separate Items)	
		L – Adjustable Long Delay Pickup (By Adjustable Rating Plug) S – Adjustable Short Delay Pickup with Fixed Short Delay Time (I^2t Response) or Adjustable Short Time (Flat Response) I – Adjustable Instantaneous Pickup by Setting Short Time Delay Time to Instantaneous G – Adjustable Ground Fault Pickup with Adjustable Ground Fault Delay (Flat Response)				Fixed	Adjustable
		LS	LSI	LSG	LSIG		
		Catalog Number				Catalog Number	
Short Time Range Short Time Delay Ground Fault Pickup Ground Fault Delay		2 – 8 x I_n — — —	2 – 8 x I_n 0 – 300 ms — —	2 – 8 x I_n — Varies by Frame 0 – 500 ms	2 – 8 x I_n 0 – 300 ms Varies by Frame 0 – 500 ms		
Type HKDB with Digitrip 310 Non-Interchangeable Trip Unit Suitable for Reverse Feed							
125	3	**HKDB3125FT33W**	**HKDB3125FT32W**	**HKDB3125FT35W**	**HKDB3125FT36W**	**1KES070T** **1KES090T** **1KES100T** **1KES125T**	70/90/100/125 **AKES125T1**
250	3	**HKDB3250FT33W**	**HKDB3250FT32W**	**HKDB3250FT35W**	**HKDB3250FT36W**	**2KES070T** **2KES090T** **2KES100T** **2KES125T** **2KES150T** **2KES160T** **2KES175T** **2KES200T** **2KES225T** **2KES250T**	125/150/200/250 **A2KES250T1**
400	3	**HKDB3400FT33W**	**HKDB3400FT32W**	**HKDB3400FT35W**	**HKDB3400FT36W**	**4KES200T** **4KES225T** **4KES250T** **4KES300T** **4KES350T** **4KES400T**	125/150/200/250 **A4KES250T1**
							200/240/260/280 **A4KES200T5**
							320/340/360/380 **A4KES300T5**

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100% Rated Types CKD and CHKD Electronic Circuit Breakers

The NEC allows the breaker to be rated at 100% of its frame size in an assembly, provided that 90°C wire is applied at the 75°C ampacity. All 100% rated circuit breakers have electronic trip units.

Table 12-180. 100% Rated Types CKD and CHKD Electronic Circuit Breakers ①

Maximum Continuous Ampere Rating at 40°C	Circuit Breaker Frame Only		Digitrip RMS 310 Trip Unit Only				Digitrip 310 Rating Plug Only			Standard Terminals Only See Page 12-107 for Optional Terminals
	Standard Interrupting Capacity 35 kAIC at 480 Vac	High Interrupting Capacity 65 kAIC at 480 Vac	Standard	Options			Ampere Rating	Fixed Rating Plug	Adjustable Rating Plug Ampere Rating	
			Adjustable Short Time Pickup with I^2t Short Delay Ramp	Independently Adjustable Short Time Pickup and Delay	Adjustable Short Time Pickup with I^2t Short Delay and Ground Fault Protection	Independently Adjustable Short Time Pickup and Delay and Ground Fault Protection				
	Catalog Number							Catalog Number		
3-Pole										
125	**CKD3400F**	**CHKD3400F**	**KES3125LS**	**KES3125LSI**	**KES3125LSG**	**KES3125LSIG**	70 90 100 110 125	**1KES070T** **1KES090T** **1KES100T** **1KES110T** **1KES125T**	70/90/100/125 **A1KES125T1**	**TA300K** ② **TA300K** ② **TA300K** ② **TA300K** ② **TA300K** ②
250	**CKD3400F**	**CHKD3400F**	**KES3250LS**	**KES3250LSI**	**KES3250LSG**	**KES3250LSIG**	70 100 125 150 160 175 200 225 250	**2KES070T** **2KES100T** **2KES125T** **2KES150T** **2KES160T** **2KES175T** **2KES200T** **2KES225T** **2KES250T**	125/150/200/225 **A2KES250T1**	**TA300K** ② **TA300K** ② **TA300K** ② **TA300K** ② **TA300K** ② **TA300K** ② **TA300K** ② **TA300K** ② **TA350K** ②
400	**CKD3400F**	**CHKD3400F**	**KES3400LS**	**KES3400LSI**	**KES3400LSG**	**KES3400LSIG**	200 225 250 300 350 400	**4KES200T** **4KES225T** **4KES250T** **4KES300T** **4KES350T** **4KES400T**	200/250/300/400 **A4KES400T1** 220/240/260/280 **A4KES200T5** 320/340/360/380 **A4KES300T5**	**TA300K** ② **TA300K** ② **TA350K** ② **TA350K** ② **TA350K** ② **3TA400K** ③

① See **Table 12-181** below for prices.
② Individually packed.
③ 3TA400K terminal kit contains one terminal for each pole and one terminal cover.

Table 12-181. 100% Rated Types CKD and CHKD Electronic Circuit Breakers Prices

Catalog Number	Price U.S. $	Catalog Number	Price U.S. $	Catalog Number	Price U.S. $	Catalog Number	Price U.S. $
A1KES125T1 **A2KES250T1** **A4KES400T1** **A4KES200T5** **A4KES300T5**		**KES3250LSG** **KES3250LSI** **KES3250LSIG** **KES3400LS** **KES3400LSG**		**1KES070T** **1KES090T** **1KES100T** **1KES110T** **1KES125T**		**2KES200T** **2KES225T** **2KES250T** **3TA400K** —	
CHKD3400F **CKD3400F** **KES3125LS** **KES3125LSG** **KES3125LSI** **KES3125LSIG** **KES3250LS**		**KES3400LSI** **KES3400LSIG** **TA300K** **TA350K** — — —		**2KES070T** **2KES100T** **2KES125T** **2KES150T** **2KES160T** **2KES175T** —		**4KES200T** **4KES225T** **4KES250T** **4KES300T** **4KES350T** **4KES400T** —	